

ELEC 872 - Replication Lab 1: LBP

First Author

Syed Taha Yeasin Ramadan
tahayeasin@queensu.ca

1 Discussion

Figure 1 shows the recognition rate of the baseline LBP face recognition system on the ORL dataset under the tested distortions (uniform brightness change, lighting ramps of increasing strength, and in-plane rotations), run on my local machine.

Figure 2 shows the results of the improved version of the baseline under the same distortions and evaluation protocol framework.

Figure 3 compares the baseline and the improved version directly.

So I've used Tan-Triggs acts like a band-pass filter. The difference of Gaussians strips away low-frequency gradients (the lighting shadow across the face) while preserving the high-frequency facial textures. So, these are the reason this come off as better improvement over the baselines.

Histogram Equalization is a global method. It spreads out contrast across the entire image. If half the face is dark and half is bright, global equalization stretches both sides equally.

The code files are upload at the LBP:
<https://github.com/yeasintaha/LBP.git>

```

PS C:\Drive_D\Queen's\Courses\ELEC 872\LBP\LBP> python baseline.py
baseline                0.975
uniform brightness      0.975
lighting ramp 0.2 0.96
lighting ramp 0.4 0.945
lighting ramp 0.6 0.895
lighting ramp 0.8 0.705
rotate 5 deg 0.97
rotate 10 deg 0.925
rotate 20 deg 0.63
PS C:\Drive_D\Queen's\Courses\ELEC 872\LBP\LBP> 

```

Figure 1: Baseline Results.

```

PS C:\Drive_D\Queen's\Courses\ELEC 872\LBP\LBP> python baseline_improve.py

Evaluating: Baseline (Raw Images)
baseline (no distortion)      0.9750
uniform brightness (0.5x + 20) 0.9750
Lighting ramp:
  ramp 0.2: 0.9600 | ramp 0.4: 0.9450 | ramp 0.6: 0.8950 | ramp 0.8: 0.7050
Rotation:
  rot 5 deg: 0.9700 | rot 10 deg: 0.9250 | rot 20 deg: 0.6300

Evaluating: Global Histogram Equalization
baseline (no distortion)      0.9800
uniform brightness (0.5x + 20) 0.9750
Lighting ramp:
  ramp 0.2: 0.9600 | ramp 0.4: 0.9500 | ramp 0.6: 0.8750 | ramp 0.8: 0.6950
Rotation:
  rot 5 deg: 0.9700 | rot 10 deg: 0.9300 | rot 20 deg: 0.6350

Evaluating: Tan-Triggs Illumination Normalization
baseline (no distortion)      0.9700
uniform brightness (0.5x + 20) 0.9700
Lighting ramp:
  ramp 0.2: 0.9700 | ramp 0.4: 0.9700 | ramp 0.6: 0.9650 | ramp 0.8: 0.9650
Rotation:
  rot 5 deg: 0.9600 | rot 10 deg: 0.9150 | rot 20 deg: 0.5250

Saved results table to: results\robustness_improvements.csv

```

Figure 2: Improved version of the improved LBP system under the same distortions as in Figure 1.

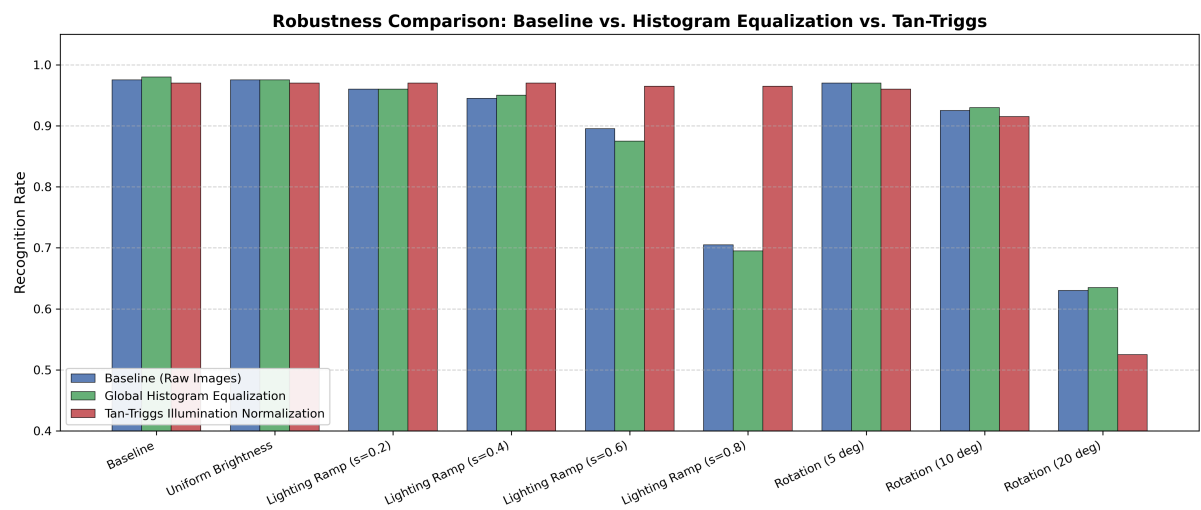


Figure 3: Robustness comparison between the baseline and the improved LBP system on the ORL dataset across all tested distortions.